The Classical Lefschetz Fixed-Point Theorem

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Abstract

The Weil conjectures sought to generalize a classical theorem from topology, the Lefschetz fixed point theorem, which lets you count the fixed points of a self-mapping of a compact space. There are many incarnations of this theorem, and we will untangle them a little bit, but we mainly focus on the case of smooth manifolds and de Rham cohomology.

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1 Background

1.1 Cohomology and integration

The Lefschetz fixed-point theorem is a result on counting fixed points that applies to many different kinds of spaces, with differing levels of regularity giving us different levels of information. It expresses information about the fixed points of a function through the induced map on (co)homology. To use the intuition that comes with differential forms, we are going to stick with smooth manifolds and de Rham cohomology. To state the theorem, let us first recall a few terms.

Definition 1.1. Suppose M is an n-dimensional smooth manifold. Then the graded algebra of alternating covariant tensors $\Omega^\bullet(M)$ is called the *de Rham* complex and its elements *differential forms*. A differential form belonging to the k-graded subalgebra $\Omega^k(M)$ is known as a *k-form*. The *exterior derivative* operator d is defined via

$$d\left(\sum_I f_I dx_I\right) = df_I dx_I$$

where df_I is the regular differential on functions. With d as the coboundary operator, $\Omega^\bullet(M)$ forms a chain complex and the cohomology of this chain complex $H^\bullet(M) = H^\bullet_{dR}(M)$ is known as the *de Rham cohomology* of M. Differential forms in the image of d are called *exact forms* and differential forms in the kernel are called *closed forms*. Hence representatives of a cohomology class are closed forms and two closed forms belong to the same cohomology class if and only if they differ by an exact form.

Differential forms exist to be integrated. We can integrate an n-form over our manifold, and with a little more machinery we can integrate k-forms over k-dimensional submanifolds. This is achieved via a pullback, which is a more general operation.

Definition 1.2. If $f\colon M \to N$ is a smooth map between smooth manifolds, and ω is a k-form on N, then we call $f^*\omega$ *the pullback of* ω *along* f which can be defined in coordinates

$$f^*\left(\sum_I g_I dy_{i_1} \wedge \cdots \wedge dyi_r\right) = \sum_I (g_I \circ f) df_{i_i} \wedge \cdots \wedge df_{i_r}$$

This operation generalizes the pullback of functions and 1-forms. Now, if S is a k-dimensional submanifold of M and ω a k-form defined in M, the inclusion map ι induces a pullback and we can integrate $\iota^*\omega$ over S, provided it has compact support. We will denote $\int_S \iota^*\omega = \int_S \omega$ when there is no confusion.

Thus, we can define integration on $\Omega^\bullet(M)$, but later we will want to have an integration operation on $H^\bullet(M)$. It is not entirely clear that we can integrate *cohomology classes*, because each class is represented by different forms which could integrate to different numbers. For n-forms, we need to see when integration over M is well defined, and for all other forms we also need to see if the pullback along the inclusion ι^* is a well-defined operator. The second is always true:

Theorem 1.3. The exterior derivative commutes with pullbacks. In particular, this means that the pullback is a well-defined operation on cohomology.

Remark 1.4. In more algebraic language, this says that the pullback on differential forms is a cochain map $\Omega^\bullet(N) \to \Omega^\bullet(M)$, which means that it induces a well-defined map on cohomology.

Now we can uniformly discuss whether integration is a well defined operation on $H^k(M)$ for any k. If our submanifold is without boundary, this is true, and we get this via Stokes' theorem.

Theorem 1.5 (Stokes' Theorem)**.** For a smooth orientable manifold M with boundary ∂M given the induced orientation,, a compactly-supported $(n-1)$-form ω satisfies

$$\int_M d\omega = \int_{\partial M} \omega$$

Corollary 1.6. Suppose M is an n-dimensional manifold and$S$ a k-dimensional compact oriented submanifold *without boundary*. Then the map $\Omega^k(M) \to \mathbb{R}$ given by $\omega \mapsto \int_S \omega$ descends to a map on cohomology $H^k(M) \to \mathbb{R}$ given by $[\omega] \mapsto \int_S \omega$

Proof. We need to show this is independent of cohomology representative. Suppose $d\alpha$ is an exact form. Then

$$\int_S d\alpha = \int_{\partial S} \alpha = \int_{\emptyset} \alpha = 0$$

Hence $\int_S \omega + d\alpha = \int_S \omega$ □

1.2 The statement of the theorem

The Lefschetz fixed-point theorem is a statement about the following quantities.

Definition 1.7. If $f\colon M \to M$ is a self map of a smooth manifold, then we denote by $\mathrm{Fix}(f)$ the set of fixed points of that map i.e. points p such that $f(p) = p$. A fixed point is said to be *isolated* if around it we can find a small enough ball such that no other fixed point sits inside the ball. For an isolated fixed point p of a smooth manifold, its *fixed-point index*, denoted $\mathrm{Ind}(f, p)$, is defined to be the degree of the map $\frac{f(x)-x}{|f(x)-x|}$ restricted to the boundary of a coordinate ball around p small enough so as to contain no other fixed points.

Remark 1.8. We will discuss the fixed point index, and alternative definitions of it, in a later section.

Theorem 1.9 (Lefschetz fixed-point theorem)**.** Let M be an n-dimensional oriented compact smooth manifold without boundary and $f\colon M \to M$ be a map with isolated fixed points. Then

$$\Lambda_f \coloneqq \sum_{i=0}^{n} (-1)^i \operatorname{Tr}(f_i^*) = \sum_{p \in \mathrm{Fix}(f)} \mathrm{Ind}(f, p)$$

where $\operatorname{Tr}(f_i^*)$ is the trace of the pullback map on the i-th cohomology group.

The quantity Λ_f is called the *Lefschetz number* of f and is well-defined because

- There are only finitely many nonzero cohomology groups on a manifold
- Cohomology groups are finite-dimensional
- The pullback is a well-defined linear map on cohomology
- This is self map, so f_i^* is an endomorphism
- The trace of an endomorphism is well-defined and basis-independent

Remark 1.10. We will be proving this theorem on closed (compact and without boundary) oriented smooth manifolds, even though many of the tools we build along the way can be extended to more general cases. To not introduce unnecessary hypotheses, we will always assume we are working with closed oriented manifolds and closed oriented submanifolds thereof. For instance, we will not distinguish between the de Rham cohomology and compactly-supported de Rham cohomology.

There are some immediate implications of this theorem. Because the Lefschetz number of a map is defined via induced maps on cohomology, it is a homotopy invariant, so somehow the right hand side is counting fixed points "upto homotopy." This is where the index of a fixed-point comes in; as we will see, if naively count the number of fixed points we do not get something that is homotopy invariant.

1.3 Intuition from the simplicial version

Why should the number of fixed point, counted "upto homotopy," have anything to do with the trace on homology at all? Some intuition comes from the following simpler version of Lefschetz theorem, applicable to the more general class of simplicial spaces.

Theorem 1.11. Suppose X is a compact simplicial complex and $f\colon X \to X$ a continuous self map. Define

$$\Lambda_f^\Delta = \sum_{i=0}^{n} (-1)^i \operatorname{Tr}(f_* : H_i(X;\mathbb{Q}) \to H_i(X;\mathbb{Q}))$$

If $\Lambda_f^\Delta \neq 0$, then f has a fixed point.

Here, note that the induced map is on *homology*, not cohomology. This homology can be singular homology or simplicial homology, because for simplicial complexes they are the same. To understand what is happening, let's recall some definitions

Definition 1.12. A *simplicial complex* is a space X that is formed by simplices (the higher-dimensional analogues of the triangle, tetrahedron, etc) such that the boundary of a simplex is also a simplex and any two simplices intersect in another simplex. We also use this term for spaces *homeomorphic* to a simplicial complex, such as a sphere. The *simplicial chain complex with coefficients in* R, denoted $C_\bullet(X;R)$, is the free module generated by these simplices; the k-dimensional simplices generate the k-th graded ring. The differential ∂ for this complex is given by sending a simplex to the *oriented* sum of its boundary simplices; then one checks that $\partial^2 = 0$. The homology of this complex (defined exactly as for de Rham cohomology, just with differentials going the other way) is called the *simplicial homology* and denoted $H_i(X;R)$. A map $f\colon K \to L$, where K and L are both simplicial complexes, is called a *simplicial map* if it sends each simplex of K to a simplex of L by a map taking vertices to vertices. For any simplicial map, the induced map on homology is obtained by seeing which simplices got sent to which other simplices and writing it down in a matrix.

Example 1.13. The sphere is a simplicial complex. Any 3D polyhedron, for instance the platonic solids, are simplicial complexes homeomorphic to the 2-sphere. Euler's formula $F + V - E = 2$ is in fact computing the Euler characteristic of the simplicial chain complex of the 2-sphere.

In the statement of the theorem, we confounded simplicial and singular homology. The induced map on singular homology can be calculated for *any* map, but for simplicial homology only for simplicial maps. Thus given a self-map, we want to approximate it using simplicial maps.

Theorem 1.14 (Simplicial approximation theorem)**.** Given a map $f\colon K \to L$ between simplicial complexes, we can approximate f arbitrarily closely with homotopic maps $f_t\colon K' \to L'$ where K' and L' are subdivisions of K and L.

Sketch of proof of the simplicial Lefschetz theorem. Assume that f has no fixed points. We show that Λ_f^Δ is zero.

1. First, we apply the simplicial approximation theorem and some cleverness to obtain a map g that is simplicial, homotopic to f, and has fixed points if and only if f does. By homotopy invariance, $\Lambda_g^\Delta = \Lambda_f^\Delta$
2. We compute the induced map $g_*\colon C_\bullet(X) \to C_\bullet(X)$. Because it has no fixed points, it sends no simplex to itself. Thus the matrices for the induced map have zeroes on the diagonal
3. We show that $\sum_{i=0}^n (-1)^i \operatorname{Tr}(g_* : H_i(X) \to H_i(X)) = \sum_{i=0}^n (-1)^i \operatorname{Tr}(g_* : C_i(X) \to C_i(X))$, hence $\Lambda_g^\Delta = 0$

□

This is a coarser theorem than the main theorem we are proving, because it only tells us the existence of a fixed point. Even so, this is all we need for some applications, as we'll see in the applications section. It has the bonus of applying to more general spaces. It *is* possible to develop a theory of indices on more general spaces, but that takes us a little too far afield. For the rest of the exposition we will focus on the de Rham cohomology case, where intersection theory gives us an excellent interpretation of the indices and the count of fixed points.

2 Proof of the Theorem

2.1 The fixed-point index and intersection theory on smooth manifolds

To understand and to prove the theorem, we will need to better understand the fixed-point index defined above. This index ultimately comes from intersection theory, so here we digress briefly and discuss the overall intersection-theoretic idea behind the Lefschetz theorem. Suppose M is a closed orientable manifold and $f\colon M \to M$ a self map with isolated fixed points. Consider the product manifold $M \times M$, and within it, the diagonal $\Delta = \{(p,p)|p \in M\}$ and the graph $\Gamma_f = \{(p, f(p))|p \in M\}$. A fixed point of f is the same as an intersection point of Δ and Γ_f, so intersection theory is the natural environment to prove this theorem.

The fixed-point index has many definitions:

Theorem 2.1. Let M be a closed orientable manifold and $f\colon M \to M$ a smooth map with isolated fixed points. If p is a fixed point of f, then the following quantities coincide where they are well defined:

(1) The fixed point index $\mathrm{Ind}(f,p)$ as defined above i.e. the Brouwer degree of $\frac{f(x)-x}{|f(x)-x|}$ on a small sphere containing p but no other fixed point

(2) The sign of the determinant of $df_p - I$, if $df_p - I$ is nonsingular. Note that df_p maps T_pM to T_pM, and that this condition is equivalent to df_p having no eigenvectors of eigenvalue ± 1. If this condition holds then we say that p is a *Lefschetz fixed point*

(3) The sign of the orientation of $T_{(p,p)}(M \times M)$ as compared to the orientation induced from the intersecting submanifolds $T_{(p,p)}\Delta$ and $T_{(p,p)}\Gamma_f$, whenever they intersect transversely. This condition is equivalent to p being a Lefschetz fixed point

(4) If $M = \mathbb{CP}^1$ and f holomorphic, the multiplicity of the zero of $f(z) - z$ at p.

(5) If V is a vector field on M and f its small-time flow, then the Brouwer degree of $\frac{v}{|v|}$ around a point where V is zero (so the flow has a fixed point). This is the Poincare-Hopf theorem, discussed below

Proof. ((1) $\iff$ (2)) If our fixed point is Lefschetz, then we approximate $\frac{f(x)-x}{|f(x)-x|}$ by its first-order approximation $\frac{A(x)-x}{|A(x)-x|}$ where A is linear, and show that their degrees are the same, and that it is the sign of determinant of $df_p - I$ i.e. whether $df_p - I$ is orientation-preserving or orientation-reversing. In fact, more is true. If p is a non-Lefschetz fixed point, we can approximate f by a family of homotopic maps f_t that are equal to f outside of a small neighborhood of p, such that all fixed point of f inside this neighborhood are Lefschetz and their indices add up to the index of p. This is the *splitting principle*.

((2) $\iff$ (3)) Choose a basis $v_1, \ldots, v_n$ for T_pM, getting bases $(v_1, v_1), \ldots, (v_n, v_n)$ for $T_{(p,p)}M \times M$ and $(v_1, df_p v_1), \ldots, (v_n, df_p v_n)$. Transform the latter into the basis $(0, (df_p - I)v_1), \ldots, (0, (df_p - I)v_n)$ and then

the former into $(v_1, 0), \ldots, (v_n, 0)$. Now when we concatenate these bases, the direct sum orientation will agree with the natural orientation of $T_{(p,p)}M \times M$ if and only if $df_p - I$ is orientation-preserving.

((1) $\iff$ (4)) The degree calculation gives us the winding number of $f(z) - z$ around a zero, which we know from complex analysis is the multiplicity of the zero. This essentially comes from approximating f as z^m. Even though we state this for $\mathbb{CP}^1$, it is true in the (noncompact) $\mathbb{C}$ too.

((1) $\iff$ (5)) See the proof below of the Poincare-Hopf theorem $\square$

Remark 2.2 (Transversality homotopy theorem)**.** There is a hierarchy of regularities of fixed points, with non-isolated fixed points being the least regular, followed by isolated, followed by the Lefschetz fixed points. It first it seems our methods only apply to Lefschetz fixed points, but in fact they are more general. Because the Lefschetz number (and as we'll see, the intersection number) is a property of a *homotopy class* of maps, it holds true for any nice map homotopic to our map. The *transversality homotopy theorem* says that we can approximate any map as closely as desired by "transverse maps" (in this case, that means that the diagonal and graph intersect transversely), so if our map has bad fixed points, we can study the maps that are close to it that have only Lefschetz fixed points. If our map has only isolated fixed points, if we perturb it slightly our non-Lefschetz fixed points will break down into Lefschetz fixed points, and the sum of their fixed-point index is the fixed-point index of the non-Lefschetz fixed point. From now on, even when we only assume isolated fixed point, we will freely allow the graph and the diagonal to be transverse to each other and go on as if all fixed points were Lefschetz

Remark 2.3 (Fixed-point index in complex manifolds)**.** When M is complex and f holomorphic, this degree is always positive. We can see this by considering the basis $\frac{\partial}{\partial x_j} + i\frac{\partial}{\partial y_j} = \frac{\partial}{\partial z_j}$. The differentials of holomorphic maps can rearrange the z_j anyhow, but they keep the x_j and y_j in relative order i.e. this is an even permutation. Thus it preserves orientation. For instance, in the 1-dimensional case the multiplicity of a zero is always positive.

It is the induced orientation formulation of the fixed point that will be most useful for us.

Definition 2.4. Let M be an n-dimensional closed oriented manifold and S, N be closed oriented submanifolds of dimensions k and $n-k$ respectively. Moreover, let S and N intersect transversely. Then by transversality they intersect in isolated points and by compactness the number of these points is finite. At each such point p, $T_pM = T_pS \oplus T_pN$ has an orientation that either agrees with or disagrees with the direct sum orientation induced by T_pS and T_pN. If they agree, define $\mathrm{Or}_M^{S,N}(p)$ to be $+1$, and -1 otherwise. Define the *intersection number* $I(S, N)$ to be

$$I(S, N) = \sum_{p \in S \cap N} \mathrm{Or}_M^{S,N}(p)$$

Theorem 2.5. The intersection number of two submanifolds is homotopy invariant. In detail, if $f_0 \colon S \to M$ and $f_1 \colon S \to M$ are two homotopic inclusion maps, then $I(f_0(S), N) = I(f_1(S), N)$, and similarly for the second slot.

Proof. See Guillemin-Pollack, pg 115. $\square$

Theorem 2.6. Suppose M is a closed orientable manifold and $f \colon M \to M$ a self map with isolated fixed points. Then

$$\mathrm{Ind}(f) := \sum_{p \in \mathrm{Fix}(p)} \mathrm{Ind}(f, p) = I_{M \times M}(\Delta, \Gamma_f)$$

and this number is homotopy-invariant

Proof. By the theorem on equivalent definitions of the index, $\mathrm{Ind}(f, p) = \mathrm{Or}_{M \times M}^{\Delta, \Gamma_f}(p)$. Then

$$\sum_{p \in \mathrm{Fix}(p)} \mathrm{Ind}(f, p) = \sum_{p \in \mathrm{Fix}(p)} \mathrm{Or}_{M \times M}^{\Delta, \Gamma_f}(p) = \sum_{p \in \Delta \cap \Gamma_f} \mathrm{Or}_{M \times M}^{\Delta, \Gamma_f}(p) = I_{M \times M}(\Delta, \Gamma_f)$$

Then homotopy invariance follows from the homotopy invariance of the intersection number. □

2.2 Poincare duality

We have reduced the problem to computing the intersection number $I_{M\times M}(\Delta, \Gamma_f)$. For this we shall make use of the following theorem

Theorem 2.7 (Intersection wedge product)**.** Let M be an n-dimensional closed oriented manifold and S, N be closed oriented submanifolds of dimensions k and $n-k$ respectively. Then there exists an $n-k$ form η_S and a k form η_N such that

$$I(S,N) = \int_M \eta_S \wedge \eta_N$$

The set of all such forms η_S (and resp for N) is a single cohomology class, which is exactly the *Poincare dual* of S, to be defined below.

To prove this theorem we will need to understand Poincare duality. There are many versions of Poincare duality, but in our circumstances (dealing only with cohomology, and everything is compact), we will use the following version: for any $0 \leq k \leq n$, there exists a pairing $\langle \cdot, \cdot \rangle$ between $H^k(M)$ and $H^{n-k}(M)$ given by

$$\langle \omega, \eta \rangle = \int_M \omega \wedge \eta$$

Poincare duality states this pairing is perfect, so every linear functional on $H^k(M)$ is represented by a class in $H^{n-k}(M)$. Now, given a k-dimensional closed oriented submanifold S of M, we have a linear functional $\int_S -$ on $H^k(M)$. This means that there exists a class η_S, called the *Poincare dual of* S, such that for every class $\omega \in H^k(M)$

$$\int_S \omega = \int_M \eta_S \wedge \omega$$

Theorem 2.8 (Poincare duality)**.** For every k-dimensional closed oriented submanifold S of M, there exists a unique $n-k$ cohomology class η_S that satisfies the condition above.

The Poincare dual can be thought of as more than just a cohomology class associated to our manifold. The idea is that since a class representing the Poincare dual expresses the action of "integrating against S," it should only be supported near S. If it was not, then we could cook up a form ω that was supported far away from S, so $\int_S \omega = 0$, but which had support near the element of our Poincare dual away from S. This idea is made precise by the

Theorem 2.9 (Localization Principle)**.** The Poincare dual of a submanifold S is unique upto cohomology. This cohomology class has representatives with support arbitrarily near S. In fact, $[\eta]$ is the Poincare dual of S if and only if for any tubular neighborhood $\mathcal{N}_S$ of S, there exists an $\alpha \in [\eta]$ such that α is supported in N_S and $\int_{\pi_{N_S}^{-1}(x)} \alpha = 1$.

Remark 2.10. A clarification on the condition $\int_{\pi_{N_S}^{-1}(x)} \alpha = 1$. The Poincare dual η is represented by $n-k$ forms, and the dimension of a fiber of the normal bundle over S is also $n-k$, so we can integrate the form over the fiber. This characterization of the Poincare dual ultimately comes from the *Thom isomorphism theorem*, which we will not get into.

One way of thinking about the Poincare dual is as a Diract-delta-type object supported only on S. This is not completely true because cohomology represents *equivalence classes* of forms. If we look at the Poincare

dual to a point, it is any compactly supported form with total integral 1 i.e. *any* form, supported anywhere, with total integral 1. The relevant part is that we can *represent* it using forms supported arbitrarily near the point. Since integration is a well-defined operation on cohomology, this interpretation will still give us the right answer.

The intersection-theoretic idea here is this: let S and N be submanifolds of complementary dimension k and $n-k$, and choose representatives η_S and η_N of their Poincare duals such that η_S is supported in a tiny tubular neighborhood around S and η_N is supported in a tiny tubular neighborhood around N. Then $\eta_S \wedge \eta_N$ is only nonzero in a tiny neighborhood around the *intersection points* of S and N!

Sketch of proof of 2.7. The idea is to make the previous paragraph solid.

1. Show the naturality of the Poincare dual i.e. if $F\colon M \to N$ is a smooth map and S a submanifold of N, $[F^*\eta_S] = [\eta_{F^{-1}(S)}]$
2. If S_1 and S_2 are submanifolds of N, use this naturality with their inclusion maps to show that $[\eta_{S_1} \wedge \eta_{S_2}] = \eta_{S_1 \cap S_2}$
3. In the complementary dimension transverse intersection case, carefully check orientations to see that around a point of intersection $\eta_{S_1 \cap S_2}$ has sign ± 1 depending on $\mathrm{Or}_N^{S_1,S_2}$

□

2.3 Computing the intersection number

Recall that our whole project of doing intersection theory was to compute the intersection of the diagonal with the graph in $M \times M$. Now we have finally reduced that problem to evaluating

$$\int_{M\times M} \eta_\Delta \times \eta_{\Gamma_f} = \int_{\Gamma_f} \eta_\Delta$$

For this, we will need to construct an explicit form representing the Poincare dual of Δ and integrate it. The Poincare dual of the diagonal lives in $H^n(M \times M)$. Fortunately, the Kunneth formula tells us the structure of $H^\bullet(M \times M)$.

Theorem 2.11 (Kunneth Formula)**.** Let M_1 and M_2be closed oriented manifolds and $M_1 \times M_2$ the product manifold, with π_1 and π_2 the projections to M_1 and M_2 respectively. Then there is an isomorphism $H^\bullet(M_1) \otimes H^\bullet(M_2) \cong H^\bullet(M_1 \times M_2)$ given by $\alpha \otimes \beta \mapsto \pi_1^*(\alpha) \wedge \pi_2^*(\beta)$.

Remark 2.12. This is another place where working in de Rham cohomology (or in general, cohomology defined over a field) comes in handy. The Kunneth formula is more complicated for integer cohomology, for instance.

Now, we choose a basis ω_i of $H^\bullet(M)$, and denote τ_i the Poincare dual basis i.e.

$$\int_M \omega_i \wedge \tau_j = \delta_{ij}$$

where we use the convention if the integrand is not of the correct dimension then the integral is zero. By Kunneth's theorem, we can pull it back to the basis $\pi_1^*\omega_i \wedge \pi_2^*\tau_j$ of $H^\bullet(M \times M)$.

Proposition 2.13. In this basis, the Poincare dual of the diagonal is

$$\eta_\Delta = \sum (-1)^{\deg \tau_i} \pi_1^*\omega_i \wedge \pi_2^*\tau_i$$

Proof. All we need to check is that it satisfies the property of the Poincare dual; uniqueness of the Poincare dual will tell us that this then is *the* Poincare dual of Δ. The Poincare dual η_Δ satisfies

$$\int_{M\times M} \eta_\Delta \wedge \alpha = \int_\Delta \alpha$$

To break this down, we choose a basis for $H^\bullet(M \times M)$ and see that our form satisfies the property on each element of the basis. This time, we pull the τ_k back along π_1 and the ω_l back along π_2, forming the basis $\pi_1^*(\tau_k) \wedge \pi_2^*(\omega_l)$ of $H^\bullet(M)$. Then,

$$\begin{aligned}
\int_{M\times M} \pi_1^*(\tau_k) \wedge \pi_2^*(\omega_l) &= \int_M (\mathrm{id}, \mathrm{id})^*(\pi_1^*(\tau_k) \wedge \pi_2^*(\omega_l)) \\
&= \int_M \tau_k \wedge \omega_l \\
&= (-1)^{\deg(\tau_k)\deg(\omega_l)} \int_M \omega_l \wedge \tau_k \\
&= (-1)^{\deg(\tau_k)\deg(\omega_l)} \delta_{kl}
\end{aligned}$$

On the other hand,

$$\begin{aligned}
&\int_{M\times M} \sum (-1)^{\deg \tau_i} \pi_1^*(\omega_i) \wedge \pi_2^*(\tau_i) \wedge \pi_1^*(\tau_k) \wedge \pi_2^*(\omega_l) \\
&= \sum_i (-1)^{\deg \tau_i} (-1)^{\deg \tau_i \deg \tau_k} (-1)^{\deg \tau_i \deg \omega_l} \int_{M\times M} \pi_1^*(\omega_i) \wedge \pi_1^*(\tau_k) \wedge \pi_2^*(\omega_l) \wedge \pi_2^*(\tau_i) \\
&= \sum_i (-1)^{\deg \tau_i} (-1)^{\deg \tau_i \deg \tau_k} (-1)^{\deg \tau_i \deg \omega_l} \left(\int_M \omega_i \wedge \tau_k\right) \left(\int_M \omega_l \wedge \tau_i\right) \\
&= \sum_i (-1)^{\deg \tau_i} (-1)^{\deg \tau_i \deg \tau_k} (-1)^{\deg \tau_i \deg \omega_l} \delta_{ik}\delta_{il} \\
&= (-1)^{\deg \tau_k} (-1)^{\deg \tau_k \deg \tau_k} (-1)^{\deg \tau_k \deg \omega_l} \delta_{kl} \\
&= (-1)^{\deg \tau_k \deg \omega_l} \delta_{kl}
\end{aligned}$$

□

Now that we have this, we can finally compute the integral.

Proposition 2.14.

$$\int_{\Gamma_f} \eta_\Delta = \Lambda_f$$

Proof.

$$\begin{aligned}
\int_{\Gamma_f} \eta_\Delta &= \int_M (\mathrm{id}, f)^*(\eta_\Delta) \\
&= \int_M (\mathrm{id}, f)^* \left(\sum_i (-1)^{\deg \tau_i} \pi_1^*(\omega_i) \wedge \pi_2^*(\tau_i)\right) \\
&= \sum_i (-1)^{\deg \tau_i} \int_M \omega_i \wedge f^*(\tau_i)
\end{aligned}$$

Recalling that τ_j is a basis for $H^\bullet(M)$, we write $f^*(\tau_i)$ as $\sum_j F_{ij}\tau_j$, then

$$\begin{aligned}\sum_i (-1)^{\deg \tau_i} \int_M \omega_i \wedge f^*(\tau_i) &= \sum_i (-1)^{\deg \tau_i} \int_M \omega_i \wedge \left(\sum_j F_{ij}\tau_j\right) \\ &= \sum_i (-1)^{\deg \tau_i} \sum_j F_{ij} \int_M \omega_i \wedge \tau_j \\ &= \sum_i (-1)^{\deg \tau_i} \sum_j F_{ij}\delta_{ij} \\ &= \sum_i (-1)^{\deg \tau_i} F_{ii}\end{aligned}$$

Divide this expression up by degree, where $n_p = \dim H^p(M)$ and F^p is the matrix for F restricted to the subspace spanned by degree p forms

$$\sum_i (-1)^{\deg \tau_i} F_{ii} = \sum_{p=0}^{n} (-1)^p \sum_{i_p=0}^{n_p} F^p_{i_p i_p} = \sum_{p=0}^{n} (-1)^p \operatorname{Tr}(F^p) = \Lambda_f$$

□

Putting it all together,

Proof of Theorem 1.9.

$$\begin{aligned}\Lambda_f &= \int_{\Gamma_f} \eta_\Delta \quad \text{(by computation)} \\ &= \int_{M\times M} \eta_\Delta \wedge \eta_{\Gamma_f} \quad \text{(by Poincare duality)} \\ &= I(\Delta, \Gamma_f) \quad \text{(by Theorem 2.7)} \\ &= \sum_{p\in \operatorname{Fix}(p)} \operatorname{Ind}(f,p) \quad \text{(by Theorem 2.6)}\end{aligned}$$

□

This completes our proof of the Lefschetz fixed point theorem. Let us recall what we did

1. We defined the index of a fixed point and showed that the sum of the indices is equal to $I(\Delta, \Gamma_f)$
2. We showed that if $S, L \subseteq M$ are appropriate for intersection theory, $I(S, L) = \int_M \eta_S \wedge \eta_L$. This step relied on Poincare duality.
3. We computed that $\int_{M\times M} \eta_\Delta \times \eta_{\Gamma_f} = \Lambda_f := \sum_{i=0}^{n} (-1)^i \operatorname{Tr}(f_i^*)$. This step relied on the Kunneth formula

3 Applications

3.1 Classical Applications

Most classical applications of the Lefschetz formula are about proving the existence of fixed points, not necessarily counting them. As such, most of them only rely on the following much coarser idea.

Theorem 3.1. If M is a closed oriented smooth manifold and $f\colon M \to M$ a self-map such that $\Lambda_f \neq 0$, then f has a fixed point.

Proof. Suppose there was no fixed point. Then the index sum in the formula is zero, so $\Lambda_f = 0$. □

Corollary 3.2. Let M be a compact orientable smooth manifold with Euler characteristic $\chi(M) \neq 0$. Then any self map $f\colon M \to M$ homotopic to the identity must have a fixed point.

Proof. First, note that if we don't have isolated fixed points then fixed points exist and we are done. Otherwise we can apply the Lefschetz theorem. Now, since f is homotopic to the identity, $\Lambda_f = \Lambda_{\mathrm{id}}$. Now, the induced map on cohomology by the identity map is the identity map, so

$$\Lambda_{\mathrm{id}} = \sum_{i=0}^{n} (-1)^i \dim H_i$$

This last quantity is the definition of the Euler characteristic. □

3.1.1 Hairy-Ball theorem

Recall that a vector field on a smooth manifold induces a flow $\phi_t\colon M \to M$ where $\phi_t(x)$ is the point you get to after following the integral curve to V through X. This exists everywhere because vector fields on a compact manifold are complete, and every ϕ_t is homotopic to the identity via $\Phi\colon I \times M \to M$, $\Phi(t, x) = \phi_t(x)$. From the discussion above, we know that every flow of a vector field has a fixed point, and we know from the theory of flows that if $\phi_t(x) = x$ for $t \in (-\epsilon, \epsilon)$ then $V(x) = 0$. However, that is not enough to get us the result we want because the Lefschetz theorem does not guarantee that ϕ_t for different t will have the *same* fixed point. We can get around this, though.

Theorem 3.3 (Hairy ball theorem)**.** Every vector field on a closed oriented manifold with nonzero Euler characteristic has zero. In particular, there are no nonvanishing vector fields on even-dimensional spheres.

Proof sketch. (1) Embed the manifold in a tubular neighborhood N in Euclidean space

(2) Construct a first-order approximation to the flow as follows: Let π be the projection map of the bundle, and $\phi_t(x) = \pi(x + vt)$

(3) See that the only fixed points of ϕ_t are the zeroes of the vector field. If $v = 0$ at a point then clearly $\phi_t(x) = x$ for a whole interval t. Conversely, if $\pi(x + vt) = x$ then $x + vt$ is in the fiber over x, so v is normal to $M \hookrightarrow N$, which can only happen when $v = 0$. Thus, for any t in a small enough neighborhood of 0, $\phi_t(x) = 0$ if and only if $V(x) = 0$

(4) Since ϕ_t is homotopic to the identity and $\chi(M) \neq 0$, ϕ_t has a fixed point, hence V has a zero.

□

3.1.2 Poincare-Hopf index theorem

A slight generalization of the above argument leads us to a more sophisticated statement about vector fields. Let ϕ_t the first-order approximation flow we constructed to prove the Hairy-ball theorem. Since it is homotopic to the identity, the Lefschetz theorem says that

$$\chi(M) = \sum_{p \in \mathrm{Fix}(\phi_t)} \mathrm{Ind}(p, \phi_t)$$

The Poincare-Hopf theorem gives us a way of computing this fixed point index without first having to construct this flow.

Theorem 3.4 (Poincare-Hopf index theorem)**.** Let V be a vector field on a closed oriented manifold, and let $Z(V)$ be the set of zeroes of the vector field. If p is an isolated zero, we define $\mathrm{Ind}(p, V)$ to be the degree of $\frac{v}{||v||}$ on a small sphere containing p but no other zero. Then, if *all* zeroes of V are isolated,

$$\chi(M) = \sum_{p \in Z(V)} \mathrm{Ind}(p, V)$$

Proof. All that this theorem is saying is that the index of p defined via $\mathrm{Ind}(p, V)$ is the same as the fixed-point index of the flow $\mathrm{Ind}(p, \phi_t)$. The result would then follow by the Lefschetz principle. Let p be an isolated zero of V. Since ϕ_t is a homotopic family, the $\mathrm{Ind}(p, \phi_t)$ is the same for all of them. Hence we can take the limit as $t \to 0$ and get

$$\begin{aligned}
\deg \frac{\phi_t(x) - x}{||\phi_t(x) - x||} &= \lim_{t \to 0} \deg \frac{\phi_t(x) - x}{||\phi_t(x) - x||} \\
&= \deg \lim_{t \to 0} \frac{\phi_t(x) - x}{||\phi_t(x) - x||} \\
&= \deg \lim_{t \to 0} \frac{\pi(x + vt) - x}{||\pi(x + vt) - x||} \\
&= \deg \frac{d\pi_x(v)}{||d\pi_x(v)||} \\
&= \deg \frac{v}{||v||}
\end{aligned}$$

where the last inequality comes from the fact that v is horizontal in the tubular neighborhood bundle. □

3.1.3 Brouwer Fixed-Point theorem

The same "coarse" fixed-point principle also gets us the Brouwer fixed-point theorem. It is important to note here that the way we have developed the Lefschetz theorem, we cannot apply it because $\mathbb{D}^n$ is a manifold with boundary. However, it still a simplicial complex, so we can use the simplicial version outlined earlier.

Theorem 3.5 (Brouwer fixed point theorem)**.** Every map from $\mathbb{D}^n$ to itself has a fixed point. Even more generally, every self-map of a compact convex subset of Euclidean space has a fixed point

Proof. The space $\mathbb{D}^n$ is contractible, hence its Euler characteristic is that of a point, 1, and every map is homotopic to the identity. Hence $\Lambda_f = 1$ so a fixed point exists. Here we are defining Λ_f via simplicial homology as in the introduction. □

3.1.4 Fundamental theorem of algebra

Applying the result on maps homotopic to the identity on complex projective space gives us the fundamental theorem of algebra.

Theorem 3.6. Every matrix $M \in \mathrm{GL}_n \mathbb{C}$ has an eigenvector

Proof. The group $\mathrm{GL}_n \mathbb{C}$ is connected, so there exists a path between any matrix and the identity. Hence any linear map on $\mathbb{C}^n \to \mathbb{C}^n$ is homotopic to the identity. If M has a zero eigenvector we are done, otherwise M descends to a map $\mathbb{CP}^{n-1}$ to itself sending a subspace v to Mv. The Euler characteristic of $\mathbb{CP}^{n-1}$ is n, so there exists a fixed point, so there exists a subspace sent to itself by M. This is exactly an eigenspace. □

Corollary 3.7 (Fundamental theorem of algebra)**.** Every nonconstant polynomial on $\mathbb{C}$ has a root in $\mathbb{C}$

Proof. Apply the preceding theorem to the companion matrix of the polynomial. □

3.1.5 Maps on projective spaces

We can carry this further and use the ring structure of $\mathbb{CP}^n$. Recall that $H^\bullet(\mathbb{CP}^n) \cong R[\alpha]/\alpha^n$, where $\deg \alpha = 2$. We saw that $f^*\alpha = r\alpha$ is the degree of a self-map f, but α generates the whole cohomology ring. Thus $\Lambda_f = 1 + r + r^2 + \ldots + r^n = \frac{r^{n+1}-1}{r-1}$. If n is even, this is never zero for integer degree (if n is odd it is zero if the degree is -1). Hence, we get that *any* map $\mathbb{CP}^{2k} \to \mathbb{CP}^{2k}$ has a fixed point, hence $\mathbb{CP}^{2k}$ has the *fixed-point property.*

Now consider $\mathbb{RP}^n$. With coefficients in $\mathbb{R}$, its cohomology is

$$H^k(\mathbb{RP}^n) = \begin{cases} \mathbb{R} & k = 0 \\ \mathbb{R} & k = n \text{ and } n \text{ is odd} \\ 0 & \text{otherwise} \end{cases}$$

If n is even, then the only nonvanishing cohomology is $H^0(\mathbb{RP}^n)$, so the Lefschetz number can never be zero. Thus $\mathbb{RP}^{2k}$ also has the fixed point property. This gets us the following

Theorem 3.8. Every polynomial $p\colon \mathbb{R} \to \mathbb{R}$ of odd-degree has a root

Proof. If p has odd degree, say $2k+1$, then its companion matrix is a linear transformation $\mathbb{R}^{2k+1} \to \mathbb{R}^{2k+1}$. If zero is an eigenvector we are done, otherwise it descends to a self map of $\mathbb{RP}^{2k}$ as in the complex case. Since this space has the fixed-point property, there exists an eigenvector, so p has a root. □

3.1.6 Fundamental theorem of algebra, again

Everything we proved above used the weakest possible version of the Lefschetz theorem, i.e. only using the fact that $\Lambda_f \neq 0 \implies \mathrm{Fix}(f) \neq \emptyset$. Let's use the whole theorem now. If we are careful with some book-keeping we can get a slightly stronger version of the fundamental theorem of algebra. Recall that a polynomial $\mathbb{C} \to \mathbb{C}$ can be extended to a map $\mathbb{CP}^1$ to $\mathbb{CP}^1$ by sending the point at infinity to itself. The cohomological degree of this map (i.e. the induced map on the top cohomology class $H^2(\mathbb{CP}^1)$ is n. [1] Since $H^0(\mathbb{CP}^1)$ is generated by by the constant function, the induced map on cohomology is the identity. Hence, $\Lambda_p = n + 1$ for any polynomial p.

Theorem 3.9 (Fundamental Theorem of Algebra)**.** If p is any degree n polynomial from $\mathbb{C} \to \mathbb{C}$, then is has exactly n roots counted with multiplicity.

Proof. Extend p to a self-map $\mathbb{CP}^1$. Consider $q(z) = p(z) + z$. Then $q(z) = z \iff p(z) = 0$, and the index of a fixed point is the multiplicity of the root of $q(z) - z = p(z)$. Hence the Lefschetz number $\Lambda_q = n + 1$ is *exactly* the total multiplicity of all the roots of p, with some extra count coming from ∞. It remains to show that the fixed-point index of ∞ is 1.

Take a chart U around ∞ that is small enough to include no other zeroes of $q(z) - z$. We compute $\tilde{q}(z)$ as the representation of q in this chart. For any $z \in U$ that is not ∞, its representation in the standard chart

[1] this statement is very close to the fundamental theorem of algebra, but in fact can be proven with a weaker idea only consider the preimages of *some* point which all have multiplicity one

$(\mathbb{C} \setminus \{\infty\})$ is $\frac{1}{z}$, so applying q to it is $q(1/z)$, and sending this back to our chart U gives us $\tilde{q}(z) = 1/q(1/z)$. If $p = z^n + a_{n-1}z^{n-1} + \ldots + a_0$, we can see that

$$\tilde{q}(z) - z = \frac{1}{p(1/z) + z} - z = \frac{z^n}{z^n p(1/z) + z^{n-1}} = \frac{-z^{n+1}p(1/z)}{z^n p(1/z) + z^{n-1}} = -z\left(\frac{z^n p(\frac{1}{z})}{z^n p(1/z) + z^{n-1}}\right)$$

This expression has a zero of order 1, so the fixed-point index at ∞ is indeed 1. □

3.2 Towards a Weil cohomology theory

Suppose X is a variety over F, which we will define as the solution set of some collection of polynomials $f_1, \ldots, f_k$ with coefficients in F. We denote its F-rational points, $X(F) = \{p \in F^n | f_1(p) = \cdots = f_k(p) = 0\}$. We can now consider the variety over $\bar{F}$, the algebraic closure of F, and denote the set of solutions $X(\bar{F})$. Suppose we wanted to count the number of points in $X(F)$ given $X(\bar{F})$. Here is one way fixed points might help.

Example 3.10. Let $X(\mathbb{R})$ be a variety over $\mathbb{R}$, and $X(\mathbb{C})$ be the extension to $\mathbb{C}$. Then some $p \in X(\mathbb{C})$ is in $X(\mathbb{R})$ if and only if $\sigma(p) = p$, where σ denotes complex conjugation.

Proof. Since σ is a field automorphism of the extension $\mathbb{C}/\mathbb{R}$, if $p \in \mathbb{C}^n$ and $\sigma(p) = p$, then in fact $p \in \mathbb{R}^n$. Thus if $p \in X(\mathbb{C})$ and $\sigma(p) = p$, in fact $p \in \mathbb{R}^n$. Since it still satisfies those polynomial equations, $p \in X(\mathbb{R})$. □

If we now think about $\mathbb{F}_q$, the finite field of order q (for q prime), then by freshman's dream we know that $\Phi_q(x) = x^q$ is the identity on $\mathbb{F}_q$ and a field automorphism for any extension $F/\mathbb{F}_q$. It is in fact true that just like σ, the *only* points it fixes are those in $\mathbb{F}_q$. Hence, if $X(\overline{\mathbb{F}_q})$ is a variety over $\overline{\mathbb{F}_q}$, its $\mathbb{F}_q$-rational points are those fixed by Φ_q. The Weil conjectures sought a cohomology theory for which we could prove a Lefschetz fixed point formula, because (under appropriate conditions of "transversality") it would give us the number of points of X in $\mathbb{F}_q$. If we look at the proof carefully, we used Poincare duality and the Kunneth formula in a big way. Hence Weil conjectured that there exists a cohomology theory on other algebraic spaces that satisfied certain axioms, such as finiteness, functoriality, Poincare duality, the Kunneth formula, and more.

4 References

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